Grade 7 Name______Answer Key__
Unit 10 Date___________________
Lesson 3 Practice Period__________________

1. A great white pelican can hold about 700 cubic inches in its bill. After flying down to try to catch a fish, the pelican has 100 cubic inches of water in his mouth. The inequality $700 \geq 100 + w$ represents this situation, with w representing the amount of space that can still be filled in the pelican's bill. What is the maximum amount of water that can be added to the bill without spilling out?

$$\begin{array}{r} 700 \geq 100 + w \\ \underline{-100 \quad -100} \\ 600 \geq w \end{array}$$

Less than or equal to 600 in^3

2. Clay County is gearing up for their annual Math Field Day competition. They need to order boxed lunches for all the proctors. Chic-Fil-A charges them $\$6.75$ per box for lunch. The budget for the event lunches is $\$225$. The inequality $\$6.75x \leq \$225$ represents the situation, with x representing the number of box lunches that can be purchased from Chic-Fil-A. What is the maximum number of boxed lunches that can be purchased?

$$\frac{6.75x}{6.75} \leq \frac{225}{6.75}$$
$$x \leq 33.\overline{3}$$

33 boxed lunches

3. The band department is raising money to purchase new uniforms by selling Mums for senior night. The team earns $\$0.75$ for each flower sold. They hope to raise at least $\$250$ from this fundraiser. This situation is represented by the inequality $\$0.75x \geq \$250$. What is the minimum number of flowers that the band must sell to reach their fundraising goal?

$$\frac{0.75x}{0.75} \geq \frac{250}{0.75}$$
$$x \geq 333.\overline{3}$$

334 flowers

4. Maximum capacity at Busch Gardens, Tampa is $33{,}000$ people. The inequality $b \leq 33{,}000$ represents this situation, with b representing the number of patrons at the park. Give an example of a solution to the inequality that is not valid in this context. Answers may vary.

-1; a negative amount of people can't visit the park.

5. Hilary is buying cupcakes from Publix to share with her 26 classmates for her birthday. The store has 18 vanilla cupcakes on the shelf. She wants to buy the remaining chocolate for the remaining cupcakes. The inequality $18 + c \geq 26$ represents the situation, with c standing for the number of cupcakes that she still needs to buy. Give an example of a solution to the inequality that is not valid in this context.

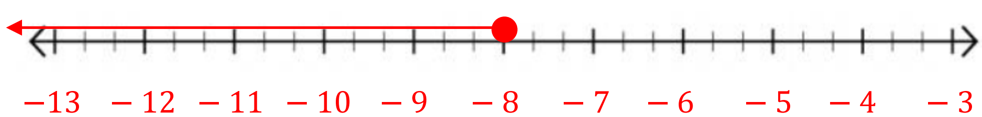
$$\begin{array}{r} 18 + c \geq 26 \\ \underline{-18 \qquad -18} \\ c \geq 8 \end{array}$$

9.5 is not a valid because you can't buy a fraction or part of a cupcake.

6. Complete the table by substituting each value of x to determine if that is a solution to each inequality.

Value of x	**What is the value of $3x$**	**Is $3x \leq -24$ true or false?**	**Solution?**
-10	-30	True	o Yes o No
-8.25	-24.75	True	o Yes o No
8.25	24.75	False	o Yes o No
10	30	False	o Yes o No

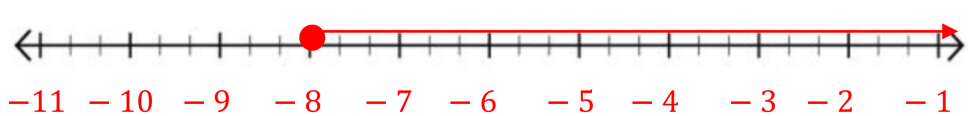
7. Graph the inequality on the number line.



8. Complete the table by substituting each value of x to determine if that is a solution to each inequality.

Value of x	**What is the value of $-3x$**	**Is $-3x \leq 24$ true or false?**	**Solution?**
-10	30	False	o Yes o No
-8.25	24.75	False	o Yes o No
8.25	-24.75	True	o Yes o No
10	-30	True	o Yes o No

9. Graph the inequality on the number line.



10. Why does context matter when solving inequalities?
Because fractional pieces and negative numbers do not apply to call contexts (ex. People)